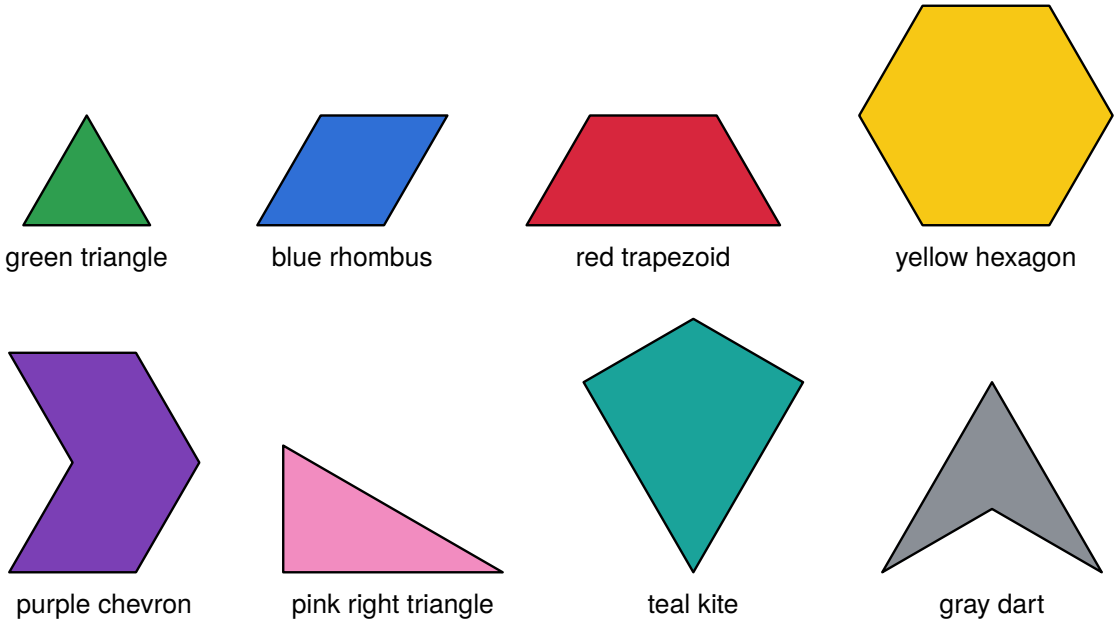
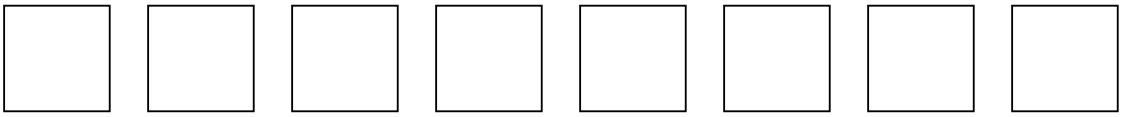
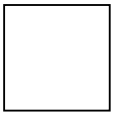
Problem 1: Your partner keeps the eight blocks behind a folder and picks one of them in secret. Ask yes-or-no questions until you are sure which block it is. Take turns, and write down how many questions each round takes.

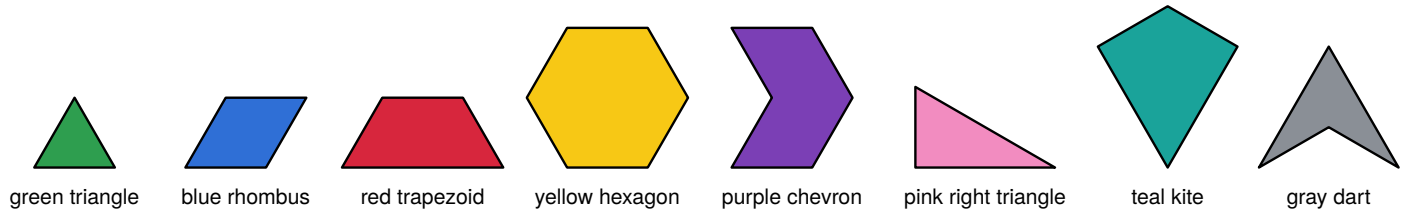




Then play some rounds where every question must name one block, such as “Is it the teal kite?” What is the most questions one of these rounds can take?

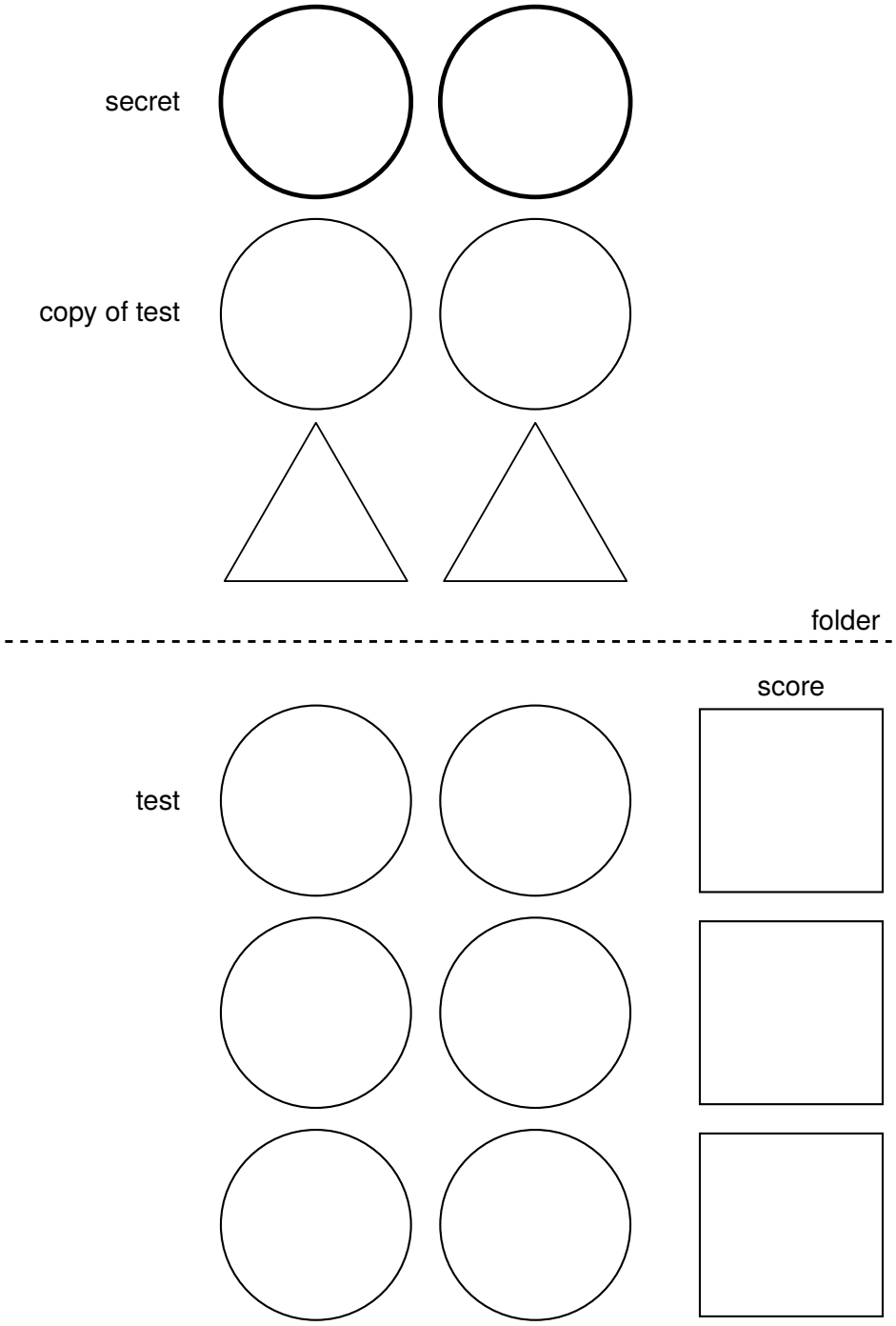


Problem 2: Find yes-or-no questions that always tell you your partner's block in three questions or fewer. Write your questions so that someone else could use them.

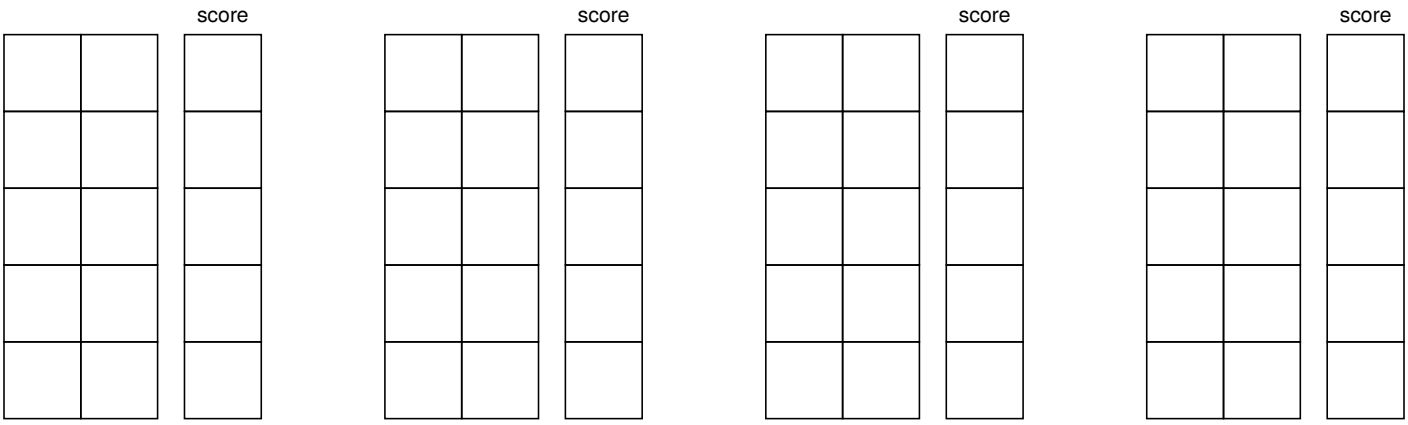


Problem 3: Can two yes-or-no questions always be enough to be sure which block it is? Explain.

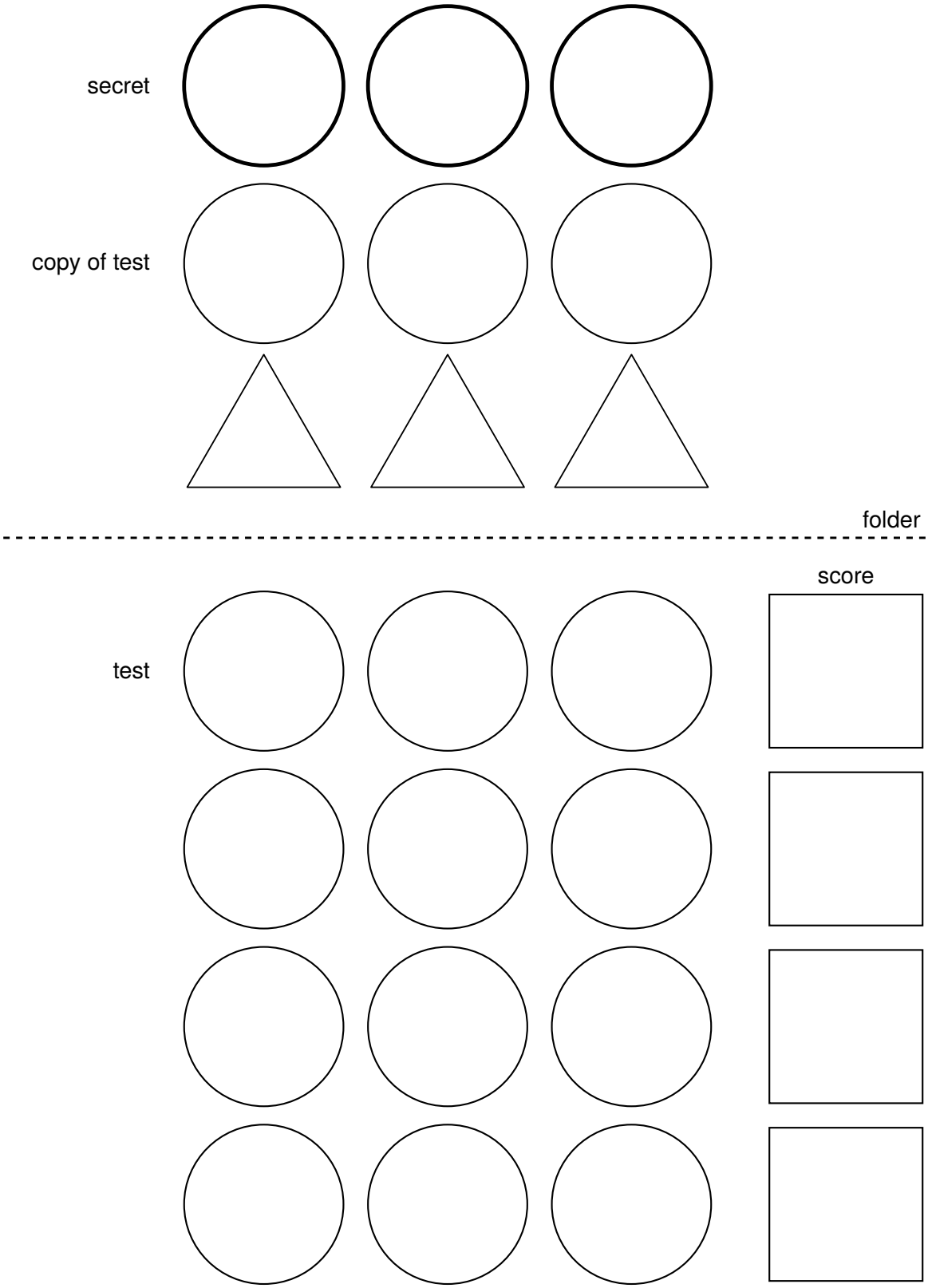
Problem 4: Play this code game with a partner, taking turns as keeper and breaker. The keeper hides a secret behind the folder: two counters, each red or yellow. The breaker builds a test in the next test row, with a red or yellow counter in every slot. The keeper copies the test just below the secret, puts a green triangle under each place where the two colors match, and says how many triangles there are. The breaker writes this score in the score box. When the breaker says what the secret is, the keeper lifts the folder and you check every score together. Find each secret with as few tests as you can.



Problem 5: Is one test always enough to be sure of a two-counter secret? Are two tests always enough? Explain.



Problem 6: Play the code game with three counters. Find each secret with as few tests as you can.



Problem 7: For each of these games, write every three-counter secret that fits all the scores. If no secret fits, explain how that could happen.

	score
R R R	2

	score
R Y R	1
Y Y R	2

	score
R Y Y	2
Y R Y	2
R R R	2

	score
R R Y	2
R Y R	2
Y R R	2

	score
R R R	3
R Y R	1